
Answer-Aware Semantic Similarity for Evaluating LLM Question Answering

Jiaxin Dong Qixiang Hua Xuqiao Ma

AIAA4051 Natural Language Processing Final Research Project

Abstract

Automatic evaluation of LLM question answering is difficult because correct answers may use different surface forms, while topically similar answers may still be factually wrong. This project studies whether embedding-space cosine similarity can serve as a correctness proxy for QA. We evaluate four benchmarks: two short-answer datasets (SCIQ, SIMPLE QUESTIONS WIKI) and two long-answer datasets (NATURAL QUESTIONS, TRUTHFULQA). A raw global cosine baseline over generated and reference factual statements performs near randomly, with AUROC between 0.482 and 0.528. We propose RefinedCosine, an answer-aware refinement pipeline that cleans prompt artifacts, extracts answer-relevant reference cores, and applies conservative lexical correction for short answers. Without changing the underlying sentence encoder, RefinedCosine improves AUROC to 0.929, 0.817, 0.760, and 0.852 on the four datasets. Ablations show that the gain mainly comes from answer-aware text construction rather than label correction alone. Additional task-specific scoring further improves robustness by using lexical-semantic hybrid scoring for short answers and claim/chunk alignment for long answers.

1 Introduction

Reliable automatic evaluation of LLM-generated answers is essential for scalable QA research. Exact match is deterministic and interpretable, but it fails when a correct answer is phrased differently from the reference. Lexical-overlap metrics such as token F1, BLEU (Papineni et al., 2002), ROUGE (Lin, 2004), and METEOR (Banerjee and Lavie, 2005) relax this constraint, yet they still measure surface similarity rather than factual correctness.

Embedding-space similarity is a natural alternative: if a sentence encoder maps semantically equivalent texts close together, then cosine similarity between a prediction and a reference may provide a continuous correctness score. However, semantic similarity and factual correctness are not identical. A wrong answer may mention the same topic or entity as the reference and still obtain a high similarity score. Long-form QA makes this harder because a single vector can hide partial correctness across multiple claims.

This work asks:

Under what text preprocessing conditions, if any, can embedding-space cosine similarity function as a reliable correctness proxy for LLM-generated question-answering outputs?

We first test a direct baseline: an instruction-tuned LLM generates answers, a statement conversion model rewrites predictions and references into standalone factual statements, an NLI model provides initial correctness labels, and cosine similarity between sentence embeddings is used as the score. This baseline performs close to random on all four datasets, showing that off-the-shelf global cosine is insufficient.

We then introduce RefinedCosine, a refined evaluation pipeline that keeps the same sentence encoder and cosine scoring but improves the comparison substrate. It removes prompt leakage, extracts answer-relevant reference cores, corrects obvious short-answer label errors with conservative lexical rules, and adds task-specific scoring for short- and long-form QA.

Contributions.

- We provide a controlled study of embedding similarity as a correctness proxy across four QA benchmarks covering short- and long-answer formats.
- We show that raw global cosine is inadequate, with AUROC only 0.482–0.528.
- We propose RefinedCosine, which improves AUROC by 0.231–0.447 without changing the underlying embedding model.
- We analyze ablations and failure cases, showing that answer-aware text construction and structured comparison are crucial for robust evaluation.

2 Related Work

Lexical evaluation metrics. Classical automatic evaluation relies on surface overlap. BLEU (Papineni et al., 2002), ROUGE (Lin, 2004), and METEOR (Banerjee and Lavie, 2005) compare generated and reference texts through n -gram overlap, recall, stemming, or synonym matching. These metrics are simple and reproducible, but they remain sensitive to surface form. This limitation motivates our study of embedding-based correctness signals.

Embedding-based generation metrics. Contextual metrics such as MoverScore (Zhao et al., 2019), BERTScore (Zhang et al., 2020), and BLEURT (Sellam et al., 2020) use pretrained representations to better handle paraphrase and semantic variation. Our work follows this direction, but the target is stricter: we evaluate binary factual correctness rather than general generation quality. This distinction matters because topical similarity can be high even when the answer is factually wrong.

Entailment and factual consistency. NLI and factuality models judge whether one text supports another. QAGS (Wang et al., 2020) and FactCC (Kryscinski et al., 2020) study factual consistency in summarization, while HHEM-2.1-Open (Saul et al., 2023) provides a practical automatic factuality labeler. We use HHEM to construct correctness labels, but our error analysis shows that NLI labels can be noisy when inputs contain prompt artifacts or when short answers are near-exact lexical matches. This motivates our conservative label correction step.

LLM factuality and long-form evaluation. Long-form factuality is difficult because a response may contain several independent claims. SelfCheckGPT (Manakul et al., 2023) estimates hallucination through output consistency, and FActScore (Min et al., 2023) decomposes long outputs into atomic facts. These works motivate our claim/chunk alignment score: instead of comparing only one global vector, we compare local prediction claims with reference chunks to better capture partial correctness.

LLM-as-a-judge evaluation. LLM-based evaluators such as G-Eval (Liu et al., 2023) and MT-Bench/Chatbot Arena (Zheng et al., 2023) can provide flexible natural-language judgments, but they introduce cost, model dependence, and calibration issues. Our goal is more lightweight and reproducible: answer-aware embedding similarity with deterministic thresholding.

Structured and geometry-aware representations. Recent work also suggests that representation geometry can affect how well models capture hierarchical semantic structure. HELM (He et al., 2025) proposes hyperbolic LLMs with mixture-of-curvature experts, while Hierarchical Mamba (Patil et al., 2025) combines Mamba-based sequence modeling with hyperbolic geometry to learn hierarchy-aware language embeddings. These methods are not directly used in our implementation. Instead, they motivate a possible future extension: our long-form failure cases indicate that a single Euclidean sentence embedding is often weak for multi-claim or hierarchical correctness, so future work could replace our current cosine scorer with hyperbolic or hierarchy-aware embeddings.

QA benchmarks. We evaluate on SCIQ (Welbl et al., 2017), SIMPLE QUESTIONS WIKI (Bordes et al., 2015), NATURAL QUESTIONS (Kwiatkowski et al., 2019), and TRUTHFULQA (Lin et al., 2022). These datasets cover short factual recall, entity-centric QA, long-form information seeking, and truthfulness-sensitive generation, allowing us to compare when embedding similarity is reliable and when it fails.

3 Problem Formulation

Let $\mathcal{D} = \{(q_i, r_i)\}_{i=1}^N$ denote a QA dataset, where q_i is a natural language question and r_i is a reference answer. For each example, an LLM generates a prediction p_i . The evaluation task is to assign a binary correctness label $\hat{y}_i \in \{0, 1\}$ to each prediction without manual grading. We measure performance in terms of ranking quality (AUROC, AUPRC) and best-threshold classification quality (Best F1) across a continuous score derived from embedding similarity.

3.1 Statement-Level Similarity

Raw answer strings are not directly comparable in many cases: a reference answer may be a bare entity span, while a model prediction may be a full declarative sentence. To reduce this representational asymmetry, we first convert each question-answer pair into a standalone factual statement via a statement conversion model $\text{Stmt}(\cdot)$:

$$\tilde{p}_i = \text{Stmt}(q_i, p_i), \quad \tilde{r}_i = \text{Stmt}(q_i, r_i).$$

Given a fixed sentence encoder $E(\cdot)$, the baseline global cosine similarity score is defined as

$$s_i = \cos(E(\tilde{p}_i), E(\tilde{r}_i)) = \frac{E(\tilde{p}_i)^\top E(\tilde{r}_i)}{\|E(\tilde{p}_i)\| \|E(\tilde{r}_i)\|}.$$

A binary correctness decision is produced by thresholding:

$$\hat{y}_i(\tau) = \mathbb{1}[s_i \geq \tau].$$

We sweep $\tau \in [0, 1]$ in increments of 0.01 and report the following evaluation metrics: accuracy, precision, recall, Best F1 (the maximum F1 score across all threshold values), AUROC, and AUPRC. AUROC captures the overall ranking quality irrespective of threshold choice, while AUPRC is particularly informative for imbalanced datasets, several of which have low positive rates in our evaluation.

3.2 Fundamental Challenges

The studied problem differs from standard semantic textual similarity in four key respects that motivate our refinement approach.

Correctness is directional. Semantic similarity is a symmetric relation, but factual correctness is not. A prediction may discuss the same topic or entity as the reference while asserting a conflicting value or relation. A high similarity score in this case constitutes a false positive.

Short answers demand near-exact matching. For knowledge-base-style questions, a single incorrect token is sufficient to render an answer factually wrong. Cosine similarity may be insufficiently sensitive to these small but semantically critical differences, particularly when surrounding context words dominate the representation.

Long answers contain multiple independent claims. A long-form prediction may correctly address some aspects of a reference while failing to support others. Representing such a prediction as a single vector conflates partial correctness with full correctness, degrading the discriminative quality of the similarity score.

Automatic labels are imperfect. NLI models used for correctness labeling can fail on exact short-answer matches that exhibit minor surface formatting differences, or on generated statements that contain prompt artifact tokens. Label quality directly constrains the maximum achievable evaluation performance, regardless of the similarity score used.

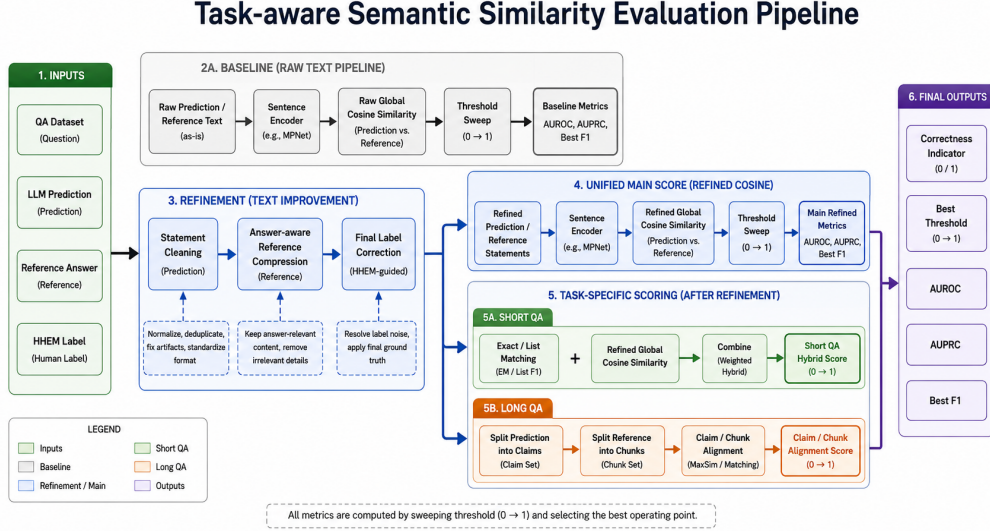


Figure 1: Overview of the evaluation pipeline.

4 Method

Figure 1 presents an overview of the complete pipeline.

4.1 Baseline Pipeline

The baseline pipeline consists of four sequential stages, each implemented with a dedicated model component.

Answer generation. For each question q_i , we deploy Llama-3.2-3B-Instruct to generate a candidate answer p_i . Short-answer datasets receive a prompt requesting a concise factual phrase or entity; long-answer datasets receive a prompt requesting a single, informative, self-contained factual sentence. Decoding is fully deterministic (temperature 0.0, top- p 1.0) to ensure reproducibility.

Statement conversion. Qwen2.5-7B-Instruct converts both p_i and r_i into standalone factual statements $\tilde{p}_i$ and $\tilde{r}_i$ by contextualizing them with the question. This step reduces the surface-form mismatch between bare entity references and full sentence predictions, making both inputs comparably structured for subsequent embedding and NLI evaluation.

Correctness labeling. HHEM-2.1-Open (Saul et al., 2023) assigns a directional entailment score to the pair $(\tilde{r}_i, \tilde{p}_i)$, where the reference statement serves as the premise and the prediction statement serves as the hypothesis. The binary baseline label is:

$$y_i^{\text{HHEM}} = \mathbb{I}[\text{HHEM}(\tilde{r}_i, \tilde{p}_i) \geq 0.5].$$

This formulation asks whether the reference supports the prediction, providing a directional factual consistency judgment.

Embedding similarity. The sentence encoder all-mpnet-base-v2 produces dense vector representations for $\tilde{p}_i$ and $\tilde{r}_i$. The global cosine between these representations constitutes the baseline correctness score s_i . This configuration is intentionally minimal: it tests whether an off-the-shelf similarity metric is already sufficient for binary correctness prediction.

4.2 RefinedCosine: Answer-Aware Refinement Pipeline

The RefinedCosine pipeline retains thresholded cosine similarity as its central mechanism but systematically improves the quality of the texts and labels that enter the evaluation.

Statement cleaning. Generated statements frequently contain copied prompt markers such as “*Question:*”, “*Answer:*”, or “*Assistant:*”, as well as repeated formatting fragments introduced by the instruction template. These artifacts are semantically irrelevant to the factual content of the answer and can distort both the embedding representation and the NLI judgment. We apply a deterministic cleaning procedure that removes known prompt leakage tokens and, when the cleaned statement is empty, falls back to a templated declarative form constructed directly from the question and raw prediction.

Reference core extraction. Long reference answers, particularly in NATURAL QUESTIONS, may contain contextual background material that extends well beyond the expected answer. Embedding the full reference can distribute representational weight across off-topic content, diluting the answer-relevant similarity signal. For long-form datasets, we decompose both the cleaned reference statement and the raw reference answer into sentence-level chunks, embed each chunk alongside the question, and select the top-two chunks by question-chunk cosine similarity. These selected chunks are concatenated in their original order to form a compact *reference core* that retains the most answer-relevant content. For short-answer datasets and TRUTHFULQA, the cleaned statement is already sufficiently concise, and the reference core defaults to the cleaned statement itself.

Refined entailment label. Following statement cleaning and reference core extraction, HHEM is re-applied to the refined pair $(\text{core}(\tilde{r}_i), \text{clean}(\tilde{p}_i))$. This produces a semantic entailment label on substantially cleaner inputs, reducing the noise introduced by prompt artifacts and irrelevant reference content.

Conservative short-answer label correction. For short-answer tasks, deterministic lexical matching can reliably identify examples where an NLI model should not reject a prediction. We compute three auxiliary scores between the raw prediction p_i and the true reference r_i : normalized exact match (EM), list-set match (ListSet), and unigram token F1. The final correctness label is defined as:

$$y_i^{\text{final}} = \begin{cases} 1, & \text{if EM}(p_i, r_i) = 1, \\ 1, & \text{if ListSet}(p_i, r_i) = 1, \\ 1, & \text{if TokenF1}(p_i, r_i) \geq 0.95, \\ y_i^{\text{refined-HHEM}}, & \text{otherwise.} \end{cases}$$

This correction rule is strictly conservative: it overrides labels to positive only under near-perfect lexical evidence, thereby addressing obvious NLI failures without introducing false positives from paraphrased but correct predictions that naturally exhibit lower lexical overlap with the reference.

Refined global cosine. The primary refined score is computed as:

$$s_i^{\text{refined}} = \cos(E(\text{clean}(\tilde{p}_i)), E(\text{core}(\tilde{r}_i))).$$

The main reported result, designated `all_refinements` in ablation analyses, evaluates s_i^{refined} against y_i^{final} .

4.3 Task-Specific Scoring Extensions

Although `all_refinements` provides a unified, dataset-agnostic evaluation score, we additionally implement and analyze task-specific scoring alternatives to determine whether the optimal metric should depend on the answer format.

Short-answer hybrid score. For short-answer QA, lexical identity and token-level precision are strong discriminative signals that complement embedding similarity. We define the hybrid score as:

$$s_i^{\text{short}} = 0.35 \cdot \text{EM}_i + 0.30 \cdot \text{TokenF1}_i + 0.15 \cdot \text{ListSet}_i + 0.20 \cdot s_i^{\text{refined}}.$$

We further evaluate a leave-one-component-out variant (`short_hybrid_without_token_f1`) that omits token F1 while retaining exact match, list-set, and refined cosine. This variant consistently outperforms the full hybrid on both short-answer datasets.

Claim/chunk alignment score. For long-answer QA, a global embedding vector may obscure partially correct responses by averaging over both supported and unsupported claims. To address this, we decompose the cleaned prediction into sentence-unit claims $\mathcal{P}$ and the reference core into sentence-unit chunks $\mathcal{R}$, then compute a pairwise cosine similarity matrix $M \in \mathbb{R}^{|\mathcal{P}| \times |\mathcal{R}|}$. We define precision and recall over this alignment:

$$P = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \max_{r \in \mathcal{R}} M_{p,r}, \quad R = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} \max_{p \in \mathcal{P}} M_{p,r}.$$

The alignment score is the harmonic mean:

$$s_i^{\text{align}} = \frac{2PR}{P+R}.$$

Precision measures whether each prediction unit is locally supported by some reference unit; recall measures whether the reference core is adequately covered by the prediction. This formulation constitutes a lightweight, embedding-based analogue of the atomic claim evaluation approach in FActScore (Min et al., 2023).

5 Experimental Setup

5.1 Datasets

Table 1 summarizes the four evaluation datasets used in this study, which collectively span short-answer factual recall, entity-centric knowledge-base QA, long-form information seeking, and truthfulness-sensitive generation.

SCIQ (Welbl et al., 2017) is a multiple-choice science dataset; we treat the correct option as the reference and generate free-form predictions, making it a single-hop factual recall task. SIMPLE QUESTIONS WIKI (Bordes et al., 2015) contains knowledge-base-style questions with single-entity answers, posing a challenging near-exact-match scenario. NATURAL QUESTIONS (Kwiatkowski et al., 2019) consists of naturally occurring Google search queries paired with Wikipedia-extracted passage answers, making it the most representative long-form setting. TRUTHFULQA (Lin et al., 2022) specifically elicits questions for which LLMs tend to reproduce common falsehoods, testing whether our metric can differentiate factually grounded from hallucinated responses.

We use the project main profile with a seeded shuffle and up to 1,000 examples per dataset; TRUTHFULQA yields 817 processed examples after filtering.

Table 1: Dataset statistics under the refined final correctness label. Positive rate denotes the fraction of examples labeled correct. Main threshold indicates the optimal decision threshold obtained by sweeping on the full evaluation set.

Dataset	Answer Type	n	Positive Rate	Main Threshold
SCIQ	Short	1000	42.5%	0.94
SIMPLE QUESTIONS WIKI	Short	1000	8.7%	0.98
NATURAL QUESTIONS	Long	1000	26.2%	0.85
TRUTHFULQA	Long	817	11.6%	0.94

5.2 Implementation Details

The complete pipeline employs four pretrained model components. Answer generation uses Llama-3.2-3B-Instruct with deterministic decoding (temperature 0.0, top- p 1.0); maximum generation lengths are 24 tokens for short answers and 128 tokens for long answers. Statement conversion uses Qwen2.5-7B-Instruct with a maximum of 192 tokens per output. Correctness labeling uses HHem-2.1-Open, a dedicated factual NLI model. Sentence embedding uses all-mpnet-base-v2 with batch size 64 and a maximum input length of 384 tokens.

Intermediate outputs—including generated predictions, statement conversions, NLI scores, and embedding vectors—are stored as JSONL, CSV, and pickle files, enabling any pipeline stage to be re-run or audited independently. Threshold sweeps are performed at step size 0.01 over $[0, 1]$.

5.3 Compared Evaluation Variants

We compare the following evaluation configurations in the main results and ablations:

- **base**: raw global cosine evaluated against original HHEM labels.
- **label_override_only**: raw global cosine evaluated against the final corrected labels, isolating the effect of label correction.
- **semantic_refinement**: refined global cosine evaluated against refined HHEM labels (without conservative overrides), isolating the effect of text cleaning and reference core extraction.
- **all_refinements**: refined global cosine evaluated against final corrected labels; this is the primary proposed method.
- **short_hybrid**: lexical-semantic hybrid scores applied to short-answer datasets.
- **claim_chunk_alignment**: local alignment score applied to long-answer datasets.

6 Results

6.1 Raw Cosine Is Near-Random

Table 2 compares the raw global cosine baseline against the full RefinedCosine pipeline. The raw baseline achieves AUROC values of 0.482 (SCIQ), 0.510 (SIMPLE QUESTIONS WIKI), 0.528 (NATURAL QUESTIONS), and 0.525 (TRUTHFULQA)—all within the range consistent with random binary classification. The metric fails to reliably rank correct predictions above incorrect ones, regardless of the answer format or domain.

This failure cannot be attributed to a single factor. Across datasets, contributing causes include: (1) generated statements that embed prompt markers alongside the factual content; (2) reference statements that encode irrelevant contextual material from long passages; (3) NLI labels that incorrectly reject near-perfect lexical matches; and (4) the inability of a single global vector to represent multi-claim long-form correctness. These findings motivate the specific interventions in RefinedCosine.

Table 2: Main results. B: raw global cosine baseline evaluated against original HHEM labels. R: RefinedCosine refined global cosine evaluated against final corrected labels. Δ : absolute improvement.

Dataset	B AUROC	R AUROC	Δ	B AUPRC	R AUPRC	Δ	B F1	R F1	Δ
SCIQ	0.482	0.929	+0.447	0.270	0.934	+0.664	0.377	0.852	+0.475
SIMPLE QUESTIONS WIKI	0.510	0.817	+0.308	0.287	0.629	+0.342	0.394	0.592	+0.198
NATURAL QUESTIONS	0.528	0.760	+0.231	0.143	0.497	+0.355	0.209	0.552	+0.343
TRUTHFULQA	0.525	0.852	+0.327	0.216	0.687	+0.470	0.236	0.632	+0.396
Short aggregate	0.492	0.918	+0.426	0.267	0.886	+0.618	0.385	0.810	+0.425
Long aggregate	0.528	0.776	+0.248	0.175	0.509	+0.334	0.219	0.498	+0.279

6.2 Refinement Produces Consistent and Substantial Improvement

The RefinedCosine pipeline improves all three evaluation metrics across all four datasets. Figure 2 visualizes the magnitude of these improvements. The largest absolute AUROC gains occur on short-answer datasets (+0.447 on SCIQ, +0.308 on SIMPLE QUESTIONS WIKI), where label correction and lexical overrides resolve a large number of false negatives arising from near-perfect lexical matches. However, the long-answer datasets also exhibit substantial improvements: NATURAL QUESTIONS AUROC increases by 0.231 and TRUTHFULQA by 0.327. This confirms that the gains are not solely attributable to the short-answer label correction mechanism.

Figure 3 provides a complementary comparison, presenting AUROC, AUPRC, and Best F1 for raw versus refined cosine, both evaluated under the same final corrected labels to isolate the contribution of text construction. The performance gap persists even under this controlled comparison, indicating that answer-aware text preprocessing provides independent benefit beyond label correction.

Figure 4 presents full ROC and precision-recall curves for the refined method across all datasets. AUPRC is an especially important diagnostic here, as SIMPLE QUESTIONS WIKI and TRUTHFULQA

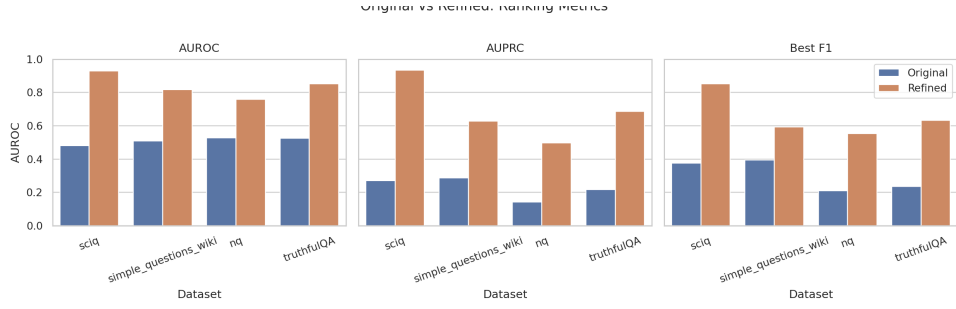


Figure 2: Before/after comparison of AUROC, AUPRC, and Best F1 across datasets. The refined RefinedCosine pipeline consistently and substantially outperforms the raw global cosine baseline on all metrics.

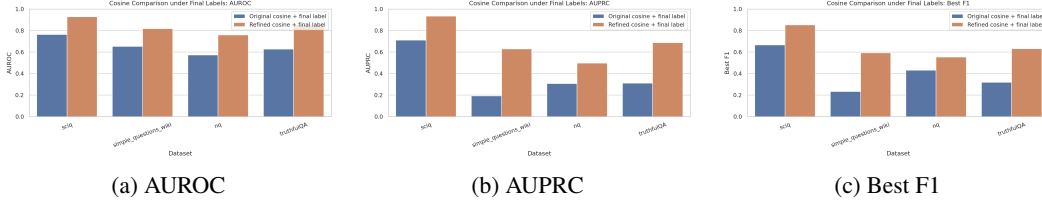


Figure 3: Comparison of raw global cosine and RefinedCosine refined cosine under the final corrected labels, isolating the effect of answer-aware text construction from label correction.

exhibit strong class imbalance (positive rates of 8.7% and 11.6%, respectively), where AUROC can appear artificially high despite poor ranking of positive examples.

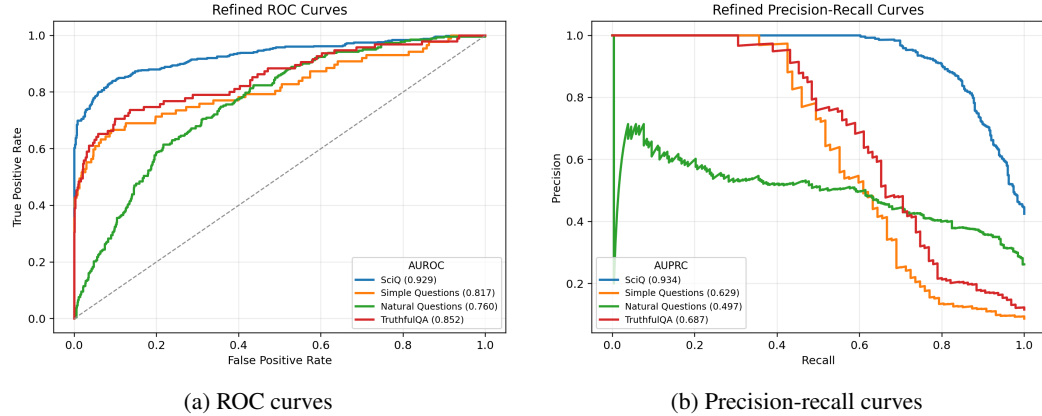


Figure 4: Refined ROC and precision-recall curves for all four datasets. The precision-recall curves reveal meaningful performance on imbalanced datasets where raw AUROC figures can be misleading.

6.3 Short-Form vs. Long-Form Discriminability

Figure 5 shows the refined cosine score distributions split by final correctness label for each dataset. Short-answer datasets exhibit substantially cleaner distributional separation between correct and incorrect predictions. Long-answer datasets show greater overlap, consistent with the expectation that partially correct responses, omitted reference facts, and unsupported peripheral claims all contribute intermediate similarity scores that are difficult to threshold cleanly.

Table 3 and Figure 6 aggregate performance by answer type. Short-form QA achieves an AUROC of 0.918 and AUPRC of 0.886, compared to 0.776 and 0.509 for long-form QA. This gap motivates the task-specific scoring extensions analyzed in the following section.

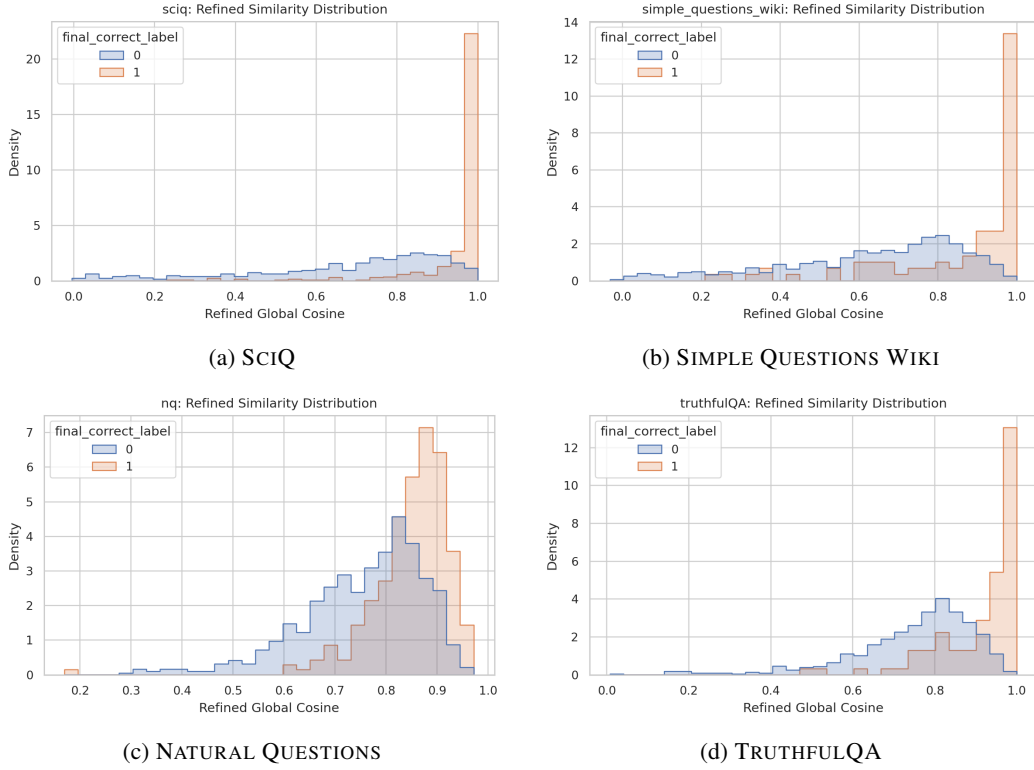


Figure 5: Distributions of refined cosine similarity scores, partitioned by final correctness label. Short-answer tasks (a, b) show substantially cleaner separation than long-answer tasks (c, d), reflecting the multi-claim nature of long-form QA correctness.

Table 3: Aggregate performance by answer format under the RefinedCosine refined evaluation. Short-form QA is pooled over SCIQ and SIMPLE QUESTIONS WIKI; long-form QA is pooled over NATURAL QUESTIONS and TRUTHFULQA.

Answer Type	n	Positive Rate	AUROC	AUPRC	Best F1
Short	2000	25.6%	0.918	0.886	0.810
Long	1817	19.6%	0.776	0.509	0.498



Figure 6: Aggregate refined evaluation performance for short-form versus long-form QA across AUROC, AUPRC, and Best F1. Short-form tasks consistently achieve higher performance owing to cleaner label separation.

7 Ablation Studies

7.1 Component Contribution Analysis

To identify which aspects of RefinedCosine drive performance, we compare four configurations that progressively introduce each component: the raw baseline, label correction in isolation, semantic text refinement in isolation, and the full pipeline. Figure 7 presents the ablation results across all three main metrics, and Table 4 lists representative numerical rows.

	AUROC				AUPRC				Best F1			
Raw cosine	0.482	0.510	0.528	0.525	0.270	0.287	0.143	0.216	0.377	0.394	0.209	0.236
Label only	0.764	0.652	0.573	0.627	0.711	0.191	0.307	0.311	0.666	0.233	0.431	0.319
Semantic refine	0.927	0.817	0.769	0.845	0.930	0.629	0.497	0.509	0.849	0.592	0.552	0.609
All refinements	0.929	0.817	0.760	0.852*	0.934	0.629	0.497	0.687*	0.852	0.592	0.552	0.632
Claim/chunk	0.929	0.825*	0.772*	0.888	0.938	0.629	0.508*	0.591	0.852	0.592	0.560*	0.638*
Short hybrid	0.915	0.795	N/A	N/A	0.920	0.442	N/A	N/A	0.828	0.439	N/A	N/A
Hybrid w/o exact	0.915	0.795	N/A	N/A	0.920	0.442	N/A	N/A	0.828	0.434	N/A	N/A
Hybrid w/o token-F1	0.933*	0.818	N/A	N/A	0.939*	0.635*	N/A	N/A	0.857*	0.604*	N/A	N/A
	SciQ	SimpleQ	NQ	TruthfulQA	SciQ	SimpleQ	NQ	TruthfulQA	SciQ	SimpleQ	NQ	TruthfulQA

* best per dataset and metric; N/A – short-only variant not applied to long QA

Figure 7: Refinement component ablation across AUROC, AUPRC, and Best F1.

Table 4: Numerical ablation results per dataset. Label correction alone provides partial improvement, but refined text construction is necessary to achieve the full performance gain.

Dataset	Variant	AUROC	AUPRC	Best F1	Interpretation
SCIQ	base	0.482	0.270	0.377	Raw cosine is near random
SCIQ	label_override_only	0.764	0.711	0.666	Corrected labels improve thresholding
SCIQ	all_refinements	0.929	0.934	0.852	Text refinement provides the main gain
SIMPLEQ	base	0.510	0.287	0.394	Raw cosine is weak
SIMPLEQ	label_override_only	0.652	0.191	0.233	Labels alone are insufficient
SIMPLEQ	all_refinements	0.817	0.629	0.592	Refined text substantially improves ranking
NQ	base	0.528	0.143	0.209	Single-vector representation is inadequate
NQ	label_override_only	0.573	0.307	0.431	Label change helps threshold calibration
NQ	all_refinements	0.760	0.497	0.552	Reference core extraction improves signal
TRUTHFULQA	base	0.525	0.216	0.236	Raw cosine is weak
TRUTHFULQA	label_override_only	0.627	0.311	0.319	Labels alone are insufficient
TRUTHFULQA	all_refinements	0.852	0.687	0.632	Clean semantic scoring is most effective

The central finding is that `label_override_only` consistently underperforms `all_refinements`. While corrected labels improve calibration for threshold-based classification—particularly on SciQ where many obvious positives are recovered—the ranking improvement measured by AUROC requires refined text embeddings. This result provides direct evidence for the claim that the quality of the text being embedded is as important as the quality of the evaluation label.

7.2 Task-Specific Best Policy

While `all_refinements` serves as the unified evaluation method for cross-dataset comparability, we additionally identify the best implemented policy per answer type. Table 5 presents these results.

For short-answer datasets, the optimal policy combines exact match, list-set match, and refined cosine, but omits token F1—a finding that suggests token overlap provides redundant signal relative to the other components when exact and list matching are already included. For long-answer datasets, claim/chunk alignment consistently matches or exceeds the unified refined cosine on all three metrics, confirming that decomposing answers into locally alignable units provides additional discriminative information beyond a single global similarity score. The exception is TRUTHFULQA AUROC and

Table 5: Performance of task-specific best scoring policies. Short-answer datasets use `short_hybrid_without_token_f1`; long-answer datasets use `claim_chunk_alignment`. Main F1 (from `all_refinements`) is provided for reference.

Dataset	Type	Task-Specific Score	AUROC	AUPRC	Best F1	Main F1
SCIQ	Short	short hybrid w/o token F1	0.933	0.939	0.857	0.852
SIMPLE QUESTIONS WIKI	Short	short hybrid w/o token F1	0.818	0.635	0.604	0.592
NATURAL QUESTIONS	Long	claim/chunk alignment	0.772	0.506	0.560	0.552
TRUTHFULQA	Long	claim/chunk alignment	0.848	0.681	0.634	0.632

AUPRC, where `all_refinements` is marginally superior (0.852 vs. 0.848 and 0.687 vs. 0.681), though the alignment score achieves the best F1.

8 Failure Analysis

A detailed examination of failure cases reveals both the limitations of the raw baseline and the residual error patterns that RefinedCosine does not fully resolve.

Short-answer label noise. A non-trivial fraction of short-answer errors in the raw baseline arise from cases where the prediction is lexically identical to the reference yet receives a low NLI score. For example, a SCIQ example with prediction “protons” and reference “protons” can be incorrectly rejected by HHEM when the statement conversion model paraphrases the two inputs into superficially different declarative forms. Conservative lexical overrides directly address this failure mode.

High similarity under factual mismatch. In science and biology domains, an incorrect answer frequently shares significant contextual vocabulary with the reference, resulting in a high cosine score despite factual disagreement. For instance, a prediction that names the wrong organelle in a cellular biology question may nonetheless be close in embedding space to the correct answer due to shared domain terminology. This pattern represents a fundamental limitation of topic-level similarity as a correctness proxy.

Prompt leakage and generation artifacts. Some generated statements incorporate prompt template fragments—particularly phrase-initial tokens such as “Question:”, “Answer:”, and “Assistant:”—that do not contribute semantic content. When such artifacts are not fully removed by statement cleaning, they can shift the embedding representation in ways that either inflate or suppress similarity scores unpredictably.

Reference dilution in long-form answers. NATURAL QUESTIONS references often include contextual sentences from Wikipedia passages that extend well beyond the core answer fact. Even with reference core extraction, some contextual material can be retained when it achieves high cosine similarity with the question, particularly when the question itself uses evaluative framing that aligns with contextual rather than answer-specific content.

Partial correctness in long-form predictions. Long-form responses may contain a correct primary claim alongside an unsupported secondary claim, or may correctly address one aspect of a multi-faceted reference while omitting others. A global similarity score cannot distinguish these cases from fully correct or fully incorrect responses at the same score level. The claim/chunk alignment score partially mitigates this through directional precision and recall decomposition, but does not resolve it entirely.

9 Discussion

The experiments support three conclusions. First, raw sentence-level cosine similarity is not a reliable correctness metric: across all datasets, the baseline is close to random. Second, the comparison substrate matters. RefinedCosine improves performance mainly by cleaning prediction statements, extracting answer-relevant reference content, and correcting obvious label noise, while keeping the same embedding model. Third, answer format should guide metric design. Short-answer QA benefits

from lexical-semantic hybrid scoring, whereas long-answer QA benefits from local claim/chunk alignment. Thus, our improvement is best understood as structured representation and task-adaptive comparison rather than a new embedding architecture.

10 Limitations

This study has four main limitations. First, all experiments use sampled subsets of up to 1,000 examples per dataset rather than full test sets because local LLM generation and statement conversion are computationally expensive. Second, the final correctness labels still depend partly on HHEM, an automatic NLI/factuality model. They should therefore be interpreted as refined automatic labels, not as definitive human judgments. Conservative lexical overrides reduce obvious short-answer errors, but borderline cases may still require expert annotation.

Third, thresholds are selected by sweeping over the same evaluation data used for reporting. This is appropriate for diagnostic analysis of score separability, but it is optimistic for deployment. A deployed system should tune thresholds on a validation split and report final performance on a held-out test split. Fourth, reference core extraction uses a simple top-two-chunk heuristic, which may miss relevant evidence in complex long-form references. Finally, RefinedCosine is validated only on QA; applying it to summarization, dialogue, or open-ended reasoning would require task-specific redesign of reference extraction and claim alignment.

11 Conclusion

This project studies when embedding-space semantic similarity can evaluate LLM QA correctness. Raw global cosine over converted factual statements is unreliable, with AUROC near 0.5 across four datasets. The main causes are prompt leakage, answer-irrelevant reference content, automatic label noise, and the weakness of one-vector comparison for multi-claim answers.

RefinedCosine addresses these issues through answer-aware statement cleaning, reference core extraction, conservative short-answer label correction, and task-specific scoring. The method improves AUROC to 0.760–0.929 without changing the underlying sentence encoder. Ablations show that the main gain comes from constructing better comparison texts, not from label correction alone. Overall, embedding similarity can be useful for QA evaluation, but only when the compared representations are explicitly grounded in the answer and adapted to the answer format.

References

- Satanjeev Banerjee and Alon Lavie. METEOR: An automatic metric for MT evaluation with improved correlation with human judgments. In *Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization*, 2005.
- Antoine Bordes, Nicolas Usunier, Sumit Chopra, and Jason Weston. Large-scale simple question answering with memory networks. *arXiv preprint arXiv:1506.02075*, 2015.
- Neil He, Rishabh Anand, Hiren Madhu, Ali Maatouk, Smita Krishnaswamy, Leandros Tassioulas, Menglin Yang, and Rex Ying. HELM: Hyperbolic large language models via mixture-of-curvature experts. *arXiv preprint arXiv:2505.24722*, 2025.
- Sarang Patil, Ashish Parmanand Pandey, Ioannis Koutis, and Mengjia Xu. Hierarchical Mamba meets hyperbolic geometry: A new paradigm for structured language embeddings. *arXiv preprint arXiv:2505.18973*, 2025.
- Wojciech Kryscinski, Bryan McCann, Caiming Xiong, and Richard Socher. Evaluating the factual consistency of abstractive text summarization. In *Proceedings of EMNLP*, 2020.
- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466, 2019.

- Chin-Yew Lin. ROUGE: A package for automatic evaluation of summaries. In *Proceedings of the Workshop on Text Summarization Branches Out*, 2004.
- Stephanie Lin, Jacob Hilton, and Owain Evans. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of ACL*, 2022.
- Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-Eval: NLG evaluation using GPT-4 with better human alignment. In *Proceedings of EMNLP*, 2023.
- Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of EMNLP*, 2023.
- Sewon Min, Kalpesh Krishna, Xinxu Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long-form text generation. In *Proceedings of EMNLP*, 2023.
- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. BLEU: A method for automatic evaluation of machine translation. In *Proceedings of ACL*, 2002.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP*, 2019.
- Vectara Inc. HHEM-2.1-Open: An open-source hallucination evaluation model. Technical report, 2023.
- Thibault Sellam, Dipanjan Das, and Ankur Parikh. BLEURT: Learning robust metrics for text generation. In *Proceedings of ACL*, 2020.
- Alex Wang, Kyunghyun Cho, and Mike Lewis. Asking and answering questions to evaluate the factual consistency of summaries. In *Proceedings of ACL*, 2020.
- Johannes Welbl, Nelson F. Liu, and Matt Gardner. Crowdsourcing multiple choice science questions. In *Proceedings of the 3rd Workshop on Noisy User-generated Text*, 2017.
- Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. BERTScore: Evaluating text generation with BERT. In *Proceedings of ICLR*, 2020.
- Wei Zhao, Maxime Peyrard, Fei Liu, Yang Gao, Christian M. Meyer, and Steffen Eger. Mover-Score: Text generation evaluating with contextualized embeddings and earth mover distance. In *Proceedings of EMNLP-IJCNLP*, 2019.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, 2023.